Adaptive Temporal Mixture of Experts for Predicting Stiffness Metrics From the Ocular Response Analyzer and Identifying Keratoconus

YUE ZHANG, SWATI PADHEE, PHILLIP T. YUHAS, CYNTHIA J. ROBERTS, AND SRINIVASAN PARTHASARATHY

• PURPOSE: To develop a machine learning–based modeling approach for extracting elastic stiffness estimates from Ocular Response Analyzer (ORA) waveforms.

• DESIGN: Prospective observational cohort study with model training and testing using a development dataset and validation with an independent validation dataset.

• METHODS: Participants were prospectively enrolled into 6 cohorts for the development dataset, including control subjects, individuals diagnosed with keratoconus, diabetes mellitus with retinopathy and without retinopathy, primary open-angle glaucoma, and ocular hypertension. An independent validation dataset comprised data for two cohorts, including healthy participants and individuals diagnosed with keratoconus. Intraocular pressure and biomechanical data were collected using the ORA and Corvis ST devices for the development dataset. An Adaptive Temporal Mixture of Experts (AT-MoE) model, incorporating long short-term memory (LSTM) networks with dynamic expert selection, was trained using ORA waveform parameters and raw signals to predict stiffness parameters (SP) from Corvis ST, including SP-A1 representing corneal stiffness, SP-HC representing scleral stiffness, and SSI representing deformation stiffness. Predicted ORA stiffness estimates were independently validated with a separate dataset of keratoconus and healthy eyes. The ORA waveform parameters, as well as applanation and pressure signals, were used as input to the machine learning models to classify eyes with keratoconus versus healthy eyes, which was evaluated with the area under the receiver operating characteristic (AUROC) curves. The primary outcome measures were the predicted ORA stiffness estimates, including corneal stiffness estimate, scleral stiffness estimate, and deformation stiffness estimate.

• RESULTS: The AT-MoE model significantly outperformed the baseline LSTM in predicting ocular stiffness estimates using ORA input data with reduction of prediction error. The AT-MoE model also resulted in improvement of keratoconus detection performance in the independent validation dataset with AUROC= 0.9512, which is similar to performance of tomographic detection approaches.

• CONCLUSIONS: The AT-MoE model provides a novel and accurate framework for deriving elastic stiffness estimates from ORA waveforms, and offers improved diagnostic capability for keratoconus compared to traditional machine learning models. This method has the potential to expand the utility of the ORA device for biomechanical assessment in clinical settings. (Am J Ophthalmol 2026;286: 196–210. © 2026 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY-NC license (http://creativecommons.org/licenses/by-nc/4.0/))

KERATOCONUS IS A PROGRESSIVE CORNEAL ECTASIA characterized by focal thinning and steepening of the cornea. Irregular astigmatism and eventual scarring lead to loss of vision and quality of life.[1] Early detection of keratoconus is critical in order to implement interventions that slow progression and preserve vision through normalization of the cornea and scar prevention.[2,3] Currently, keratoconus is diagnosed by clinical examination in conjunction with topographic or tomographic imaging of the cornea to detect focal thinning and compensatory steepening. This repetitive process leads to the proposed cyclic biomechanical decompensation of the cornea that is initiated by localized weakening at the location where the cone subsequently develops.[4] Thus, clinical biomechanical assessment of corneal elasticity can supplement imaging approaches to allow for the earliest detection.[5,6]

Currently, clinical evaluation of corneal elastic stiffness parameters is made using the Corvis ST. Although vari-

AJO.com Supplemental Material available at AJO.com.

Cynthia J. Roberts and Srinivasan Parthasarathy are co–senior authors and co–corresponding authors, with Cynthia J. Roberts handling biomechanical and Ocular Response Analyzer content and Srinivasan Parthasarathy handling content regarding machine learning.

Accepted for publication February 18, 2026.

From the Department of Computer Science and Engineering (Y.Z., Sw.P., Sr.P.), The Ohio State University, Columbus, Ohio, USA; College of Optometry (P.T.Y.), The Ohio State University, Columbus, Ohio, USA; Department of Ophthalmology and Visual Sciences (C.J.R.), The Ohio State University Wexner Medical Center, Columbus, Ohio, USA; Department of Biomedical Engineering (C.J.R.), College of Engineering, The Ohio State University, Columbus, Ohio, USA

Inquiries to Cynthia J. Roberts, The Ohio State University, Columbus, Ohio, USA; e-mail: roberts.8@osu.edu

Inquiries to Srinivasan Parthasarathy, The Ohio State University, Columbus, Ohio, USA; e-mail: srini@cse.ohio-state.edu

0002-9394/$36.00
https://doi.org/10.1016/j.ajo.2026.02.028

ous diagnostic indices have been developed to optimize the diagnostic capabilities of this device,[7] it is not commonly found in primary eye care clinics, where keratoconus is often first detected and diagnosed. Moreover, the biomechanical assessment software does not have approval from the United States Food and Drug Administration (FDA), limiting its use to measurements of intraocular pressure (IOP) and central corneal thickness (CCT) in the United States. Thus, there is a need to develop metrics of corneal elasticity in an FDA-approved device.

The Ocular Response Analyzer (ORA) is a dynamic bidirectional air-puff tonometer that currently provides two measurements of IOP and corneal hysteresis (CH), a viscoelastic biomechanical parameter that quantifies the ability of the cornea to dissipate energy.[8] Although CH is lower in eyes with keratoconus compared to healthy eyes, substantial overlap between the two groups limits its diagnostic potential.[9] However, parameters that describe the pressure–applanation waveforms may provide additional biomechanical information about the cornea.[10] These waveform parameters are altered in glaucoma[11] and are blunted in keratoconic eyes with a central cone.[12] Their diagnostic utility to detect keratoconus, particularly early disease, remains unclear, in part because interpretation of these complex waveforms requires advanced analytical tools capable of capturing both short-term fluctuations, such as the rapid applanation peaks, and long-term trends, such as the overall shape across the entire air-puff, which is influenced by scleral support and global ocular response. Therefore, advanced analytical techniques are needed to generate stiffness estimates from ORA waveforms that can be used to supplement keratoconus detection before vision impairment occurs.

Traditional machine learning models, such as long short-term memory (LSTM) networks,[13] have shown promise in capturing temporal dependencies in sequential data. LSTMs are designed to address the vanishing gradient problem, making them effective for learning from sequences. However, they often have difficulty modeling diverse temporal dependencies across different scales, as they typically learn a fixed set of representations that are not easily specialized for different parts of the sequence. Recent advances in biomedical and ophthalmic imaging analysis have demonstrated the potential of deep learning frameworks to overcome similar limitations in complex data domains.[14-17] Mixture of Experts (MoE) models provide a modular learning paradigm in which multiple expert networks are trained to specialize on different parts of the input space, with a gating network determining how to combine their outputs. This approach has been widely explored for the classification of biomedical and ophthalmic data, particularly where spatial or temporal heterogeneity poses challenges for traditional monolithic models. Early model-based frameworks demonstrated the effectiveness of expert-driven strategies in visualizing classification decisions for patient diagnosis and spatial modeling of corneal shape.[18-21] These works highlighted the value of decomposing complex spatial patterns into interpretable subcomponents, thereby improving classification accuracy in corneal imaging and videokeratography data. Inspired by these advances, the Adaptive Temporal Mixture of Experts (AT-MoE) model is designed to handle complex sequential data by leveraging modularity and dynamic routing based on the temporal context of the data. This approach allows the model to efficiently capture diverse temporal patterns across multiple timescales, such as short-term fluctuations in corneal deformation and long-term changes in corneal and scleral stiffness.

The purpose of this study was to generate elastic stiffness estimates based on ORA waveform parameters and pressure–applanation waveforms to predict Corvis ST stiffness parameters by developing a novel AT-MoE model to combine the temporal modeling capabilities of LSTM networks with the modular and sparse routing principles of MoE. The generated ORA stiffness estimates will be evaluated in a second independent validation dataset to classify individuals with keratoconus versus healthy subjects.

METHODS

Subjects in the existing development dataset in this cohort modeling study had been prospectively enrolled at The Ohio State University (OSU) Wexner Medical Center under an approved OSU Institutional Review Board protocol, and signed informed consent with Health Insurance Portability and Accountability Act (HIPPA) compliance prior to enrollment. Subjects in the existing validation dataset had been prospectively enrolled at OSU College of Optometry under a separate approved OSU Institutional Review Board protocol, and signed informed consent with HIPPA compliance prior to enrollment. Acquisition protocols differed, with the development dataset embedded within a comprehensive multi-condition research workflow, whereas the validation dataset followed a streamlined clinical protocol. ORA devices were independently maintained and operated by different clinical personnel. These differences introduce natural variability and support the external generalizability of the model. Both protocols were conducted in accordance with the tenets of the Declaration of Helsinki for research on human subjects. Both existing datasets were part of separate larger investigations of ocular biomechanics.

• **DEVELOPMENT DATASET:** The development dataset comprised subjects collected prospectively across 6 cohorts: normal, keratoconus, diabetes without retinopathy, diabetes with retinopathy, ocular hypertension, and glaucoma. These cohorts included a variety of ocular conditions, providing a comprehensive representation of different levels of corneal stiffness and ocular pathologies. This broad range from the lowest stiffness in keratoconus to the highest stiffness in ocular hypertension was essential to

TABLE 1. Demographic and Biomechanical Characteristics of Cohorts in Development Dataset.

Development Dataset Cohorts	n (Eyes)	n (Subjects)	Age (y)	IOPcc (mm Hg)	SP-A1 (mm Hg/ mm)	SP-HC (mm Hg/ mm)	SSI	CH (mm Hg)
Normal controls	397	198	39 ± 14	15 ± 3	122 ± 18	14.6 ± 3.9	1.11 ± 0.2	10.6 ± 1.5
Keratoconus	87	54	35 ± 12	14 ± 3	83 ± 22	9.4 ± 3.1	0.88 ± 0.2	8.2 ± 1.5
Diabetes	147	75	49 ± 14	15 ± 3	123 ± 15	14.9 ± 3.2	1.19 ± 0.19	10.8 ± 1.7
Diabetes with retinopathy	146	78	51 ± 14	16 ± 4	128 ± 16	17.1 ± 4.8	1.23 ± 0.22	11.6 ± 2.0
Glaucoma	101	53	61 ± 13	17 ± 4	126 ± 18	15.2 ± 4.4	1.22 ± 0.22	9.5 ± 1.9
Ocular hypertension	83	44	62 ± 17	21 ± 5	143 ± 17	23.0 ± 7.1	1.43 ± 0.27	10.3 ± 2.0

CH = corneal hysteresis; IOPcc = corneal-compensated intraocular pressure; SP-A1 = stiffness parameter at first applanation; SP-HC = stiffness parameter at highest concavity; SSI = stress–strain index.

Values are presented as mean ± SD.

ensure the robustness and generalizability of the model in various clinical scenarios. Inclusion criteria across cohorts included a clear cornea, the ability and willingness to comply with the study protocol, and an age of 18 years or older. Exclusion criteria across cohorts included the following: any comorbidity in the pathologic cohorts; nonintact epithelium; pregnancy; less than 12 weeks postpartum or from cessation of breast feeding; nystagmus; previous ocular surgery, except cataract extraction greater than 3 months prior to the date of enrollment; systemic disease that causes defects in collagen, such as Marfan syndrome; and for the normal cohort, a history of ocular disease or trauma.

Cohort eligibility and demographic data

The cohorts were as follows:

- Normal (NRL): Subjects with normal ocular health, providing baseline data for comparison with the other groups.
- Keratoconus (KCN): Individuals with a diagnosis of keratoconus with clinical signs, including at least one of reduced corneal thickness, steepening, Fleischer ring, Vogt striae, or scissoring. The focal weakening that generates focal thinning and steepening of the cornea in keratoconus made this an important group in which to study the relationship between corneal stiffness and ocular parameters.[22]
- Diabetes (DIA): Individuals diagnosed with diabetes, but limited to those without retinopathy. Participants included both type 1 and type 2 diabetes. Diabetes can affect corneal biomechanics, because tissue stiffening occurs in the presence of hyperglycemia.[23]
- Diabetic retinopathy (DR): Subjects diagnosed with diabetic retinopathy. Participants included both type 1 and type 2 diabetes, but were limited to those with nonproliferative diabetic retinopathy of mild to moderate grade in the absence of macular edema. Diabetic retinopathy may influence scleral stiffness and other ocular parameters.[24]
- Ocular hypertension (OHT): Individuals in this cohort had elevated intraocular pressure but no signs of glaucomatous damage. This group was relevant for studying corneal and scleral stiffness changes that may protect the optic nerve from glaucomatous damage.
- Glaucoma (GLA): Subjects with diagnosed primary open-angle glaucoma, a condition associated with progressive optic nerve damage and loss of visual function. Glaucoma patients often exhibit altered corneal and scleral stiffness, which is also affected by topical prostaglandin analogues as one of the pressure-lowering medications, making them a critical group for modeling stiffness metrics.[25,26]

All subjects with both Corvis ST data and ORA data were extracted from the larger dataset for a total of 961 eyes of 502 subjects in 6 cohorts. Demographic and biomechanical data for the development dataset are given in Table 1.

Devices

Corvis ST (Oculus): The Corvis ST is an advanced dynamic tonometer that uses high-speed imaging of the cornea deformation response to an air pressure pulse with consistent spatial and temporal magnitude profiles. Data exported included the biomechanical deformation response parameters of Stiffness Parameter at First Applanation (SP-A1), shown to represent corneal response; Stiffness Parameter at Highest Concavity (SP-HC), shown to represent scleral response[27]; and StressStrain Index (SSI),[28] which was developed via finite element modeling and represents deformation stiffness response. These parameters are important for understanding the relationship between intraocular pressure and ocular response, which includes both the cornea and sclera.[29]

Ocular Response Analyzer (Reichert Technologies): The ORA is used to assess corneal biomechanical response to an air pressure pulse that is customized so that an eye with lower IOP receives a lower-magnitude pulse and an eye with higher IOP receives a higher-magnitude pulse. This air-puff strategy is one feature that differentiates the ORA from the Corvis ST. A second feature that differentiates them is the electro-optical detection system used by the ORA to track corneal displacement, explained in the next section. The ORA data in the development dataset of the current study is a second-generation device (G2). Data were exported from the ORA G2 examination with the highest waveform score of the three examinations performed, and included corneal compensated intraocular pressure (IOPcc), corneal hysteresis (CH), waveform parameters, and both pressure/applanation signals, to be described in the next section.[30]

ORA waveform parameters

The ORA produces two signals by projecting an infared (IR) beam onto the cornea and collecting the diffuse reflection from the corneal surface. One signal represents the air pressure magnitude versus time, and the other represents the photons collected by the IR photodetector versus time, which spikes at both inward and outward applanation events due to a mirror-like reflection of the IR light that maximizes the light captured by the aligned detector. This second signal is called the applanation response curve; it tracks the spatial deformation information by the amount of light collected as a function of the angle of the light impinging on the detector as the corneal surface is displaced. The magnitude of the applanation signal reaches a local minimum between the two rapid applanation events and represents the maximum corneal displacement. A series of 44 waveform parameters are exported that correspond to various aspects of the corneal deformation response in the recorded signal. The waveform parameters describe the shape and characteristics of the pressure-response waveforms, for example, the heights and widths of the two applanation peaks shown in Figure 1.[12]

Data splitting

Those subjects who had both ORA data and Corvis ST data ultimately populated the development dataset. The total dataset of 502 subjects was split into training and testing subsets by subject. Any subject included in the training set was excluded from the testing set. The training set consisted of 80% of the data, whereas the remaining 20% was reserved for testing. Training, testing, and validation sets were strictly separated, ensuring no overlap of eyes from a given individual across partitions. Feature selection and normalization statistics were computed exclusively on the training set and subsequently applied to test and validation sets without recalculation, thereby eliminating the risk of data leakage.

ORA stiffness estimates

- Corneal stiffness estimate (CSe): This metric corresponds to the Stiffness Parameter at First Applanation (SP-A1) from the Corvis ST. CSe reflects the resistance of the cornea to deformation from the initial convex shape to the flattened state at applanation. It is considered a key indicator of corneal stiffness and is relevant in conditions such as keratoconus with low corneal stiffness and diabetes with high corneal stiffness.
- Scleral stiffness estimate (SSe): This metric corresponds to the Stiffness Parameter at Highest Concavity (SP-HC) from the Corvis ST. Scleral stiffness reflects the resistance of the sclera to aqueous displacement during the concave phase of deformation from applanation to the limit of corneal concavity when the sclera is fully engaged. Scleral stiffness also influences the overall biomechanical response of the ocular shell to IOP and is particularly important in the study of glaucoma and ocular hypertension.
- Deformation stiffness estimate (DSe): This metric corresponds to the Stress–Strain Index (SSI) from the Corvis ST, which uses finite element modeling of corneal deformation to the highest concavity in order to determine corneal stiffness. DSe provides useful insights into the cornea's deformation response to internal and external pressures, especially in diseases that affect corneal structure.

• **DATA PREPROCESSING:** The preprocessing of the dataset was a crucial step to ensure the quality and effectiveness of the model. The waveform parameters, which include several time-series measurements from the ORA, were subjected to normalization techniques to standardize the data and to remove any inherent biases due to differences in scale or range.

Normalization of waveform parameters

To standardize the waveform parameters across all subjects and cohorts, each parameter was normalized using max–min scaling. This technique transforms the values of each feature to a [0, 1] range based on the minimum and maximum values observed for that specific parameter across the dataset. This ensures that each feature contributes equally to the model's learning process, thereby preventing parameters with larger scales from dominating the model's performance.

The formula for max–min scaling is as follows:

$$X_{\text{norm}} = \frac{X - \min(X)}{\max(X) - \min(X)} \quad (1)$$

where X is the original value of a parameter, and X_{norm} is the normalized value.

Normalization of pressure and applanation curves

The pressure and applanation curves, which represent time-series data obtained from the ORA, were also normalized independently for each subject. This normalization was necessary because these curves can have varying magnitudes between subjects, potentially distorting the learning process if not properly adjusted.

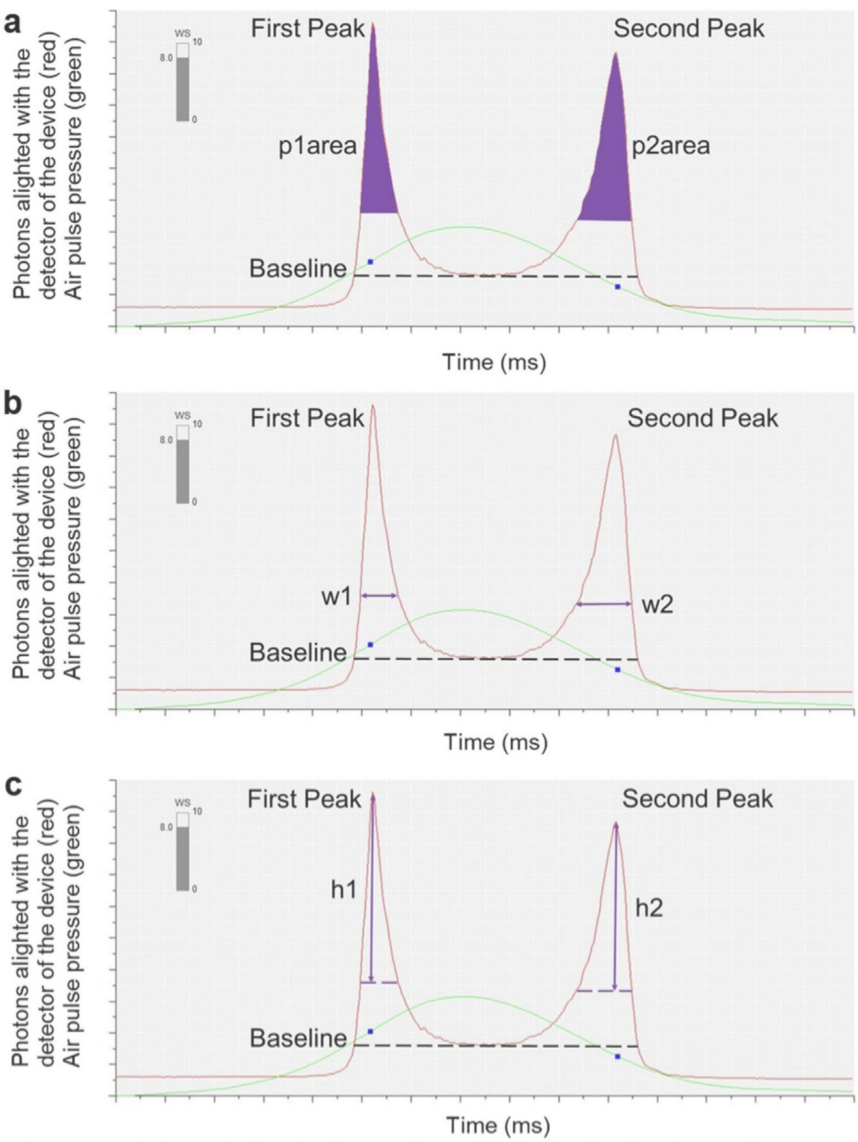

FIGURE 1. Representative pressure–applanation waveforms from the Ocular Response Analyzer (ORA). (a) Area under the first (p1area) and second (p2area) peaks, (b) the width of the first (w1) and second (w2) peaks, and (c) the height of the first (h1) and second (h2) peaks are indicated in purple. Green traces represent air pressure, and red traces represent the number of photons reflected off the corneal surface and captured by the infrared light sensor. Adapted from Yuhas et al (2024).[12]

Each pressure and applanation curve was normalized to the range [0, 1] by using the following max–min scaling procedure:

$$X_{\mathrm{norm}}(t) = \frac{X(t) - \min(X)}{\max(X) - \min(X)} \quad (2)$$

where $X(t)$ represents the value at time step t in the curve, and $X_{\mathrm{norm}}(t)$ is the normalized value for that specific time point.

• **FEATURE SELECTION:** To select the top waveform parameters most relevant to the prediction of the stiffness estimates, a carefully designed data preprocessing pipeline was used to extract the most relevant waveform parameters for the prediction of three stiffness values: scleral stiffness estimate (SSe), corneal stiffness estimate (CSe), and deformation stiffness estimate (DSe). This preprocessing pipeline combined statistical, machine learning, and optimization techniques to construct subsets of top features tailored to each stiffness value. The selected top waveform parameters included the following:

IOPg, CRF, IOPcc, p2area, p2area1, w21, CH, path1, path2, w11.

Hybrid feature selection approach

The original dataset consisted of 44 waveform parameters. Selecting a compact and informative feature set required addressing two key challenges: namely, identifying parameters that are jointly relevant to all three stiffness estimates, and avoiding redundancy among highly correlated waveform descriptors. Therefore, a hybrid feature selection approach was developed to identify a single common subset of waveform parameters used for predicting all three stiffness targets.

Multi-target statistical relevance estimation: For each waveform parameter $\mathcal{X}_i$, its dependence on each stiffness estimate $Y_k \in \{\text{SSe, CSe, DSe}\}$ was quantified using mutual information (MI), which captures both linear and nonlinear relationships:

$$MI(\mathcal{X}_i, Y_k) = \iint p(\mathcal{X}_i, Y_k) \log\left(\frac{p(\mathcal{X}_i, Y_k)}{p(\mathcal{X}_i)p(Y_k)}\right) d\mathcal{X}_i dY_k \quad (3)$$

To obtain a single relevance score per waveform parameter that reflects its overall association with all stiffness estimates, MI values were aggregated across targets:

$$\text{Relevance}(\mathcal{X}_i) = \frac{1}{3}\sum_{k=1}^{3} MI(\mathcal{X}_i, Y_k) \quad (4)$$

This multi-target relevance score favors parameters that consistently capture biomechanical information across corneal, scleral, and deformation stiffness.

Feature diversity maximization using optimization: To reduce redundancy among selected parameters, feature diversity was enforced using a maximal relevance–minimal redundancy (MRMR) criterion. Redundancy between two waveform parameters $\mathcal{X}_i$ and $\mathcal{X}_j$ were quantified using cosine similarity:

$$\text{Redundancy}(\mathcal{X}_i, \mathcal{X}_j) = \frac{\mathcal{X}_i^{\top}\mathcal{X}_j}{\|\mathcal{X}_i\|\|\mathcal{X}_j\|} \quad (5)$$

Feature selection was formulated as the following optimization problem:

$$\max_{S \subseteq F, |S|=k}\left(\sum_{\mathcal{X}_i \in S}\text{Relevance}(\mathcal{X}_i) - \lambda \sum_{\mathcal{X}_i, \mathcal{X}_j \in S}\text{Redundancy}(\mathcal{X}_i, \mathcal{X}_j)\right), \quad (6)$$

where F denotes the full set of waveform parameters, S is the selected feature subset, $k = 10$, and λ controls the trade-off between relevance and redundancy.

The optimization was solved using a greedy forward-selection strategy, in which features were iteratively added to S based on their marginal contribution to the objective. This procedure resulted in a single common set of 10 waveform parameters that was used as input for all stiffness prediction tasks.

Normalization and sequence construction

After feature selection, the selected waveform parameters for each stiffness value were normalized using z-score normalization:

$$\hat{\mathcal{X}}_i = \frac{\mathcal{X}_i - \mu(\mathcal{X}_i)}{\sigma(\mathcal{X}_i)} \quad (7)$$

where $\mu(\mathcal{X}_i)$ and $\sigma(\mathcal{X}_i)$ are the mean and standard deviation of parameter $\mathcal{X}_i$ across the training data.

• **ARCHITECTURE:** The AT-MoE architecture, shown as Figure 2, operates on the selected features generated during preprocessing. By tailoring the feature subsets to each stiffness estimate, the model is capable of capturing stiffness-specific temporal dynamics, reducing computational overhead, and enhancing predictive accuracy. The AT-MoE architecture is designed to efficiently handle sequential data by incorporating three primary components: (1) LSTM-based experts, (2) a time-sensitive gating network, and (3) a sparse routing mechanism with multi-scale temporal fusion. Each component plays a crucial role in enabling the model to learn from complex temporal patterns in the data, while also ensuring computational efficiency. In the AT-MoE architecture, each expert is an LSTM model, shown as Figure 3 designed to specialize in a specific temporal aspect of the input data, such as capturing short-term fluctuations or long-term trends. The waveform parameters are a crucial part of the input.

A detailed description of the Architecture is provided in the Supplemental Material.

• **TRAINING PROCEDURE:** The AT-MoE model is trained with a composite loss function that ensures both global optimization and expert specialization.

Model training

The model is trained on the training dataset over multiple epochs, where each epoch consists of the following steps:

Forward pass: During the forward pass, the input sequence $\mathbf{X} \in \mathbb{R}^{B \times T \times F}$ is passed through the LSTM-based experts. Each expert is configured with distinct hyperparameters, ensuring diversity in their learned representations. The attention-based gating network dynamically computes the importance of each expert based on the input features, selecting the top- k experts. The outputs of the selected experts are aggregated, and the final prediction outputs for the three stiffness values (SSe, CSe, and DSe) and cohort label are computed using dedicated, fully connected layers. The gating weights $\mathbf{G} \in \mathbb{R}^{B \times E}$, where B is the batch size and E is the number of experts, are also computed during this step.

Task-specific loss calculation: The primary task-specific loss $\mathcal{L}_{\text{task}}$ is computed as a combination of losses for the four tasks:

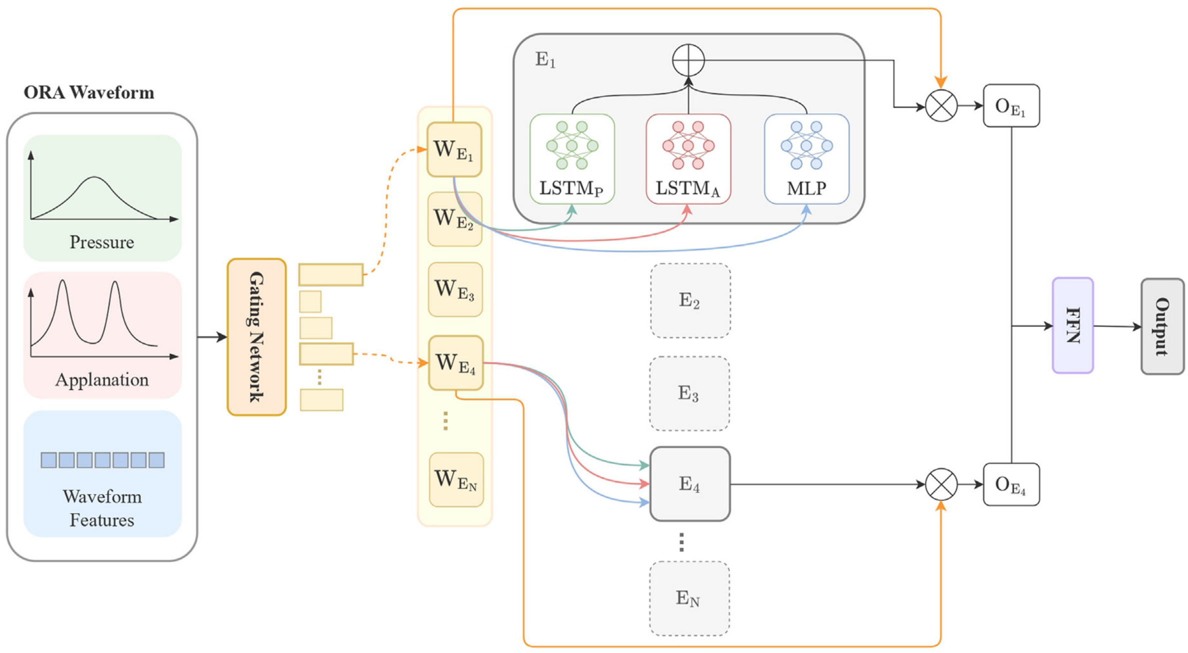

FIGURE 2. Adaptive Temporal Mixture of Experts (AT-MoE) model architecture. FFN = feed-forward network; MLP = multi-layer perceptron.

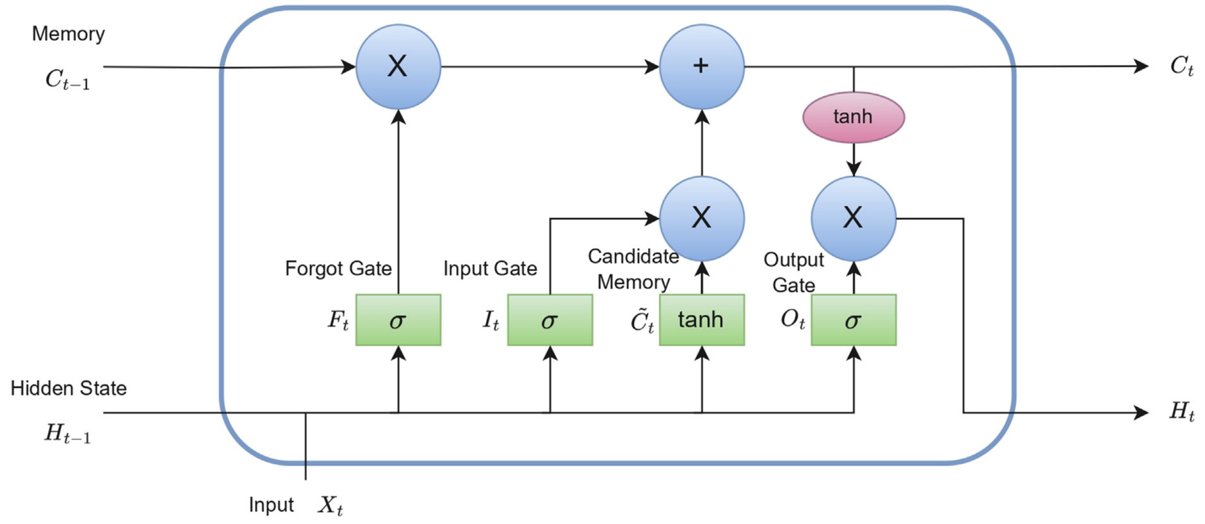

FIGURE 3. Long short-term memory (LSTM) model architecture.

- *Regression losses (CSe, SSe, DSe)*: The mean squared error (MSE) is used for the regression tasks:

$$\mathcal{L}_{\text{reg}} = \sum_{i=1}^{N} \Big(\left(Y_{\text{CSe},i} - \hat{Y}_{\text{CSe},i}\right)^2 + \left(Y_{\text{SSe},i} - \hat{Y}_{\text{SSe},i}\right)^2 + \left(Y_{\text{DSe},i} - \hat{Y}_{\text{DSe},i}\right)^2 \Big) \quad (8)$$

where N is the number of instances in the batch, and Y and $\hat{Y}$ represent the true and predicted values, respectively.

- Classification loss (cohort label): binary cross-entropy (BCE) loss is used for the classification task:

$$\mathcal{L}_{\text{class}} = -\frac{1}{N} \sum_{i=1}^{N} \left(Y_{\text{Group},i} \log\left(\hat{Y}_{\text{Group},i}\right) + \left(1 - Y_{\text{Group},i}\right) \log\left(1 - \hat{Y}_{\text{Group},i}\right)\right) \quad (9)$$

The total task-specific loss is then:

$$\mathcal{L}_{\text{task}} = \mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{class}} \tag{10}$$

Balance loss calculation: To encourage balanced utilization of the experts, the balance loss $\mathcal{L}_{\text{balance}}$ is defined as the variance of the mean gating weights across experts:

$$\mathcal{L}_{\text{balance}} = \frac{1}{E}\sum_{j=1}^{E}\left(\bar{g}_j - \frac{1}{E}\right)^2 \tag{11}$$

where E is the number of experts, and $\bar{g}_j$ is the mean gating weight for the j-th expert across all instances in the batch. This loss penalizes deviations from uniform expert utilization.

Total loss calculation: The total loss is computed as a weighted sum of the task-specific loss and the balance loss:

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \alpha\mathcal{L}_{\text{balance}} \tag{12}$$

where α is a hyperparameter that controls the importance of the balance loss.

Backpropagation and optimization: The gradients of the total loss $\mathcal{L}$ with respect to the model parameters are computed using backpropagation. The model parameters are then updated using a gradient-based optimization algorithm (e.g., Adam) to minimize the total loss.

Cross-validation strategy: To ensure robust evaluation, K-fold cross-validation was used, where the dataset was split at the subject level and each fold was used once for validation while the remaining folds were used for training.

Early stopping: To prevent overfitting, early stopping was applied. The model's performance on a validation set was monitored, and training was halted if the validation loss did not improve for a specified number of epochs. The best-performing model (based on validation loss) was saved during this process. To ensure that the large performance gains were not driven by overfitting, the training and validation loss curves were monitored, which showed stable and parallel convergence throughout training, shown in Supplemental Figure S1. The composite loss function, including task-specific loss and a balance regularization term, discouraged expert collapse and forced distributed learning across the expert pool. Sparse expert routing effectively limited the model capacity per sample, reducing memorization risk. Early stopping based on validation loss further prevented over-optimization. Together these measures ensured that the observed performance gains reflected genuine learning rather than overfitting.

Evaluation

After training, the model was evaluated on a held-out test set. The evaluation metrics included the following:

Mean squared error (MSE):
The MSE was calculated as the average of the squared differences between the predicted stiffness values $\hat{Y}$ and the true stiffness values Y:

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}\left(Y_i - \hat{Y}_i\right)^2 \tag{13}$$

Clinically, MSE reflects the overall deviation of predicted tissue stiffness from true values, with larger errors penalized more heavily. This is important when occasional large mispredictions could lead to incorrect assessment of disease severity or treatment decisions.

Mean absolute error (MAE):
The MAE was computed as the average of the absolute differences between the predicted and true stiffness values:

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|Y_i - \hat{Y}_i\right| \tag{14}$$

MAE provides the average magnitude of prediction errors in the same units as the stiffness values, giving clinicians an intuitive measure of typical model error without over-penalizing outliers.

Root mean squared error (RMSE):
The RMSE is the square root of the MSE:

$$\text{RMSE} = \sqrt{\text{MSE}} \tag{15}$$

RMSE represents the average error magnitude in the same units as the target values and gives higher weight to larger deviations. Clinically, it highlights cases in which large prediction errors occur, which may be critical for patient management decisions.

• **INDEPENDENT VALIDATION DATASET:** To assess the generalizability and performance of the model, a second independent validation dataset was used that consisted of biomechanical measurements from a first-generation (G1) ORA, including IOPcc, CH, and waveform parameters, as well as pressure and applanation signals. Two cohorts had been prospectively enrolled as described elsewhere,[31] one with a diagnosis of keratoconus, and the other consisting of healthy controls. Demographic and biomechanical data of the two cohorts are given in Table 2. Topographic data were also acquired with the E300 (Medmont International). The maximum anterior tangential curvature in a 2-mm diameter circle (Cspot) was determined with custom software. Mean, standard deviation, and range of Cspot were 53.0 ± 6.6 D (42.2 − 73.2 D). From the Medmont, flat K was reported as 47.1 ± 7.1D (38.4 − 83.3 D), and steep K as 50.8 ± 8.5D (41.8 − 91.2D).

TABLE 2. Demographic and Biomechanical Characteristics of the Validation Cohort.

Validation cohort	n (Eyes)	n (Subjects)	Age (y)	IOPcc (mmHg)	CH (mm Hg)	% Female
Normal controls	290	145	30 ± 17	14.2 ± 3.4	10.9 ± 1.8	61% Female
Keratoconus	80	44	29 ± 13	12.5 ± 2.8	8.6 ± 1.6	27% Female

CH = corneal hysteresis; IOPcc = corneal-compensated intraocular pressure.

TABLE 3. Ocular Response Analyzer Stiffness Estimates in Testing Split, 20% of Development Dataset.

Development Cohorts: 20% Testing Split by Subject	n (eyes)	n (Subjects)	Corneal Stiffness Estimate	Scleral Stiffness Estimate	Deformation Stiffness Estimate
Normal controls	80	40	126 ± 20	15.0 ± 6.0	1.11 ± 0.27
Keratoconus	18	10	88 ± 26	9.7 ± 4.4	0.8 ± 0.26
Diabetes	30	15	121 ± 23	13.4 ± 5.0	1.13 ± 0.29
Diabetes with retinopathy	30	15	131 ± 23	16.7 ± 6.5	1.21 ± 0.25
Glaucoma	19	10	124 ± 18	15.5 ± 6.9	1.14 ± 0.21
Ocular hypertension	18	9	136 ± 23	21.3 ± 8.5	1.49 ± 0.29

TABLE 4. Stiffness Prediction on the Testing Cohort Using a Single Long Short-Term Memory Model.

	MSE (95% CI)	RMSE (95% CI)	MAE (95% CI)
Corneal stiffness estimate	0.0308 (0.0250-0.0368)	0.1754 (0.1581-0.1919)	0.1411 (0.1255-0.1570)
Scleral stiffness estimate	0.0298 (0.0243-0.0357)	0.1726 (0.1560-0.1889)	0.1386 (0.1237-0.1529)
Deformation stiffness estimate	0.0344 (0.0278-0.0414)	0.1853 (0.1668-0.2034)	0.1518 (0.1362-0.1682)

CI = confidence interval; MAE = mean absolute error; MSE = mean squared error; RMSE = root mean squared error.

RESULTS

The ORA stiffness estimates for Cse, SSe, and Dse are given in Table 3 for the training split of the development dataset. Note that these will not necessarily be similar to those in the full development dataset, because the subjects included are only a subset. The predicted stiffness estimates versus the corresponding Ground Truth Stiffness Parameters from the Corvis ST are plotted with the $x = y$ unity line in Figures 4A, B, and C.

To evaluate the effectiveness of the proposed Adaptive Temporal Mixture of Experts (AT-MoE) model, we compared its performance with a baseline single LSTM model on the task of stiffness estimate prediction. Quantitative results are summarized in Tables 4 and 5. Across all three stiffness metrics—corneal stiffness estimate (CSe), scleral stiffness estimate (SSe), and deformation stiffness estimate (DSe)—the AT-MoE model demonstrated substantial improvements in prediction accuracy.

For CSe, AT-MoE reduced the mean squared error (MSE) from 0.0308 to 0.0065, which corresponds to a relative improvement of approximately 79%. The root mean squared error (RMSE) dropped from 0.1754 to 0.0806, and the mean absolute error (MAE) decreased from 0.1411 to 0.0583. These reductions indicate not only fewer large prediction errors but also a tighter concentration around the true values.

Similarly, for SSe, the MSE improved from 0.0298 to 0.0109, with the RMSE reduced from 0.1726 to 0.1044, and the MAE from 0.1386 to 0.0864. DSt followed the same trend, with the MSE decreasing from 0.0344 to 0.0188, RMSE from 0.1853 to 0.1371, and MAE from 0.1518 to 0.0975. These consistent reductions across all metrics and targets highlight the AT-MoE model's ability to model complex and diverse biomechanical patterns in sequential data more effectively than the single LSTM baseline.

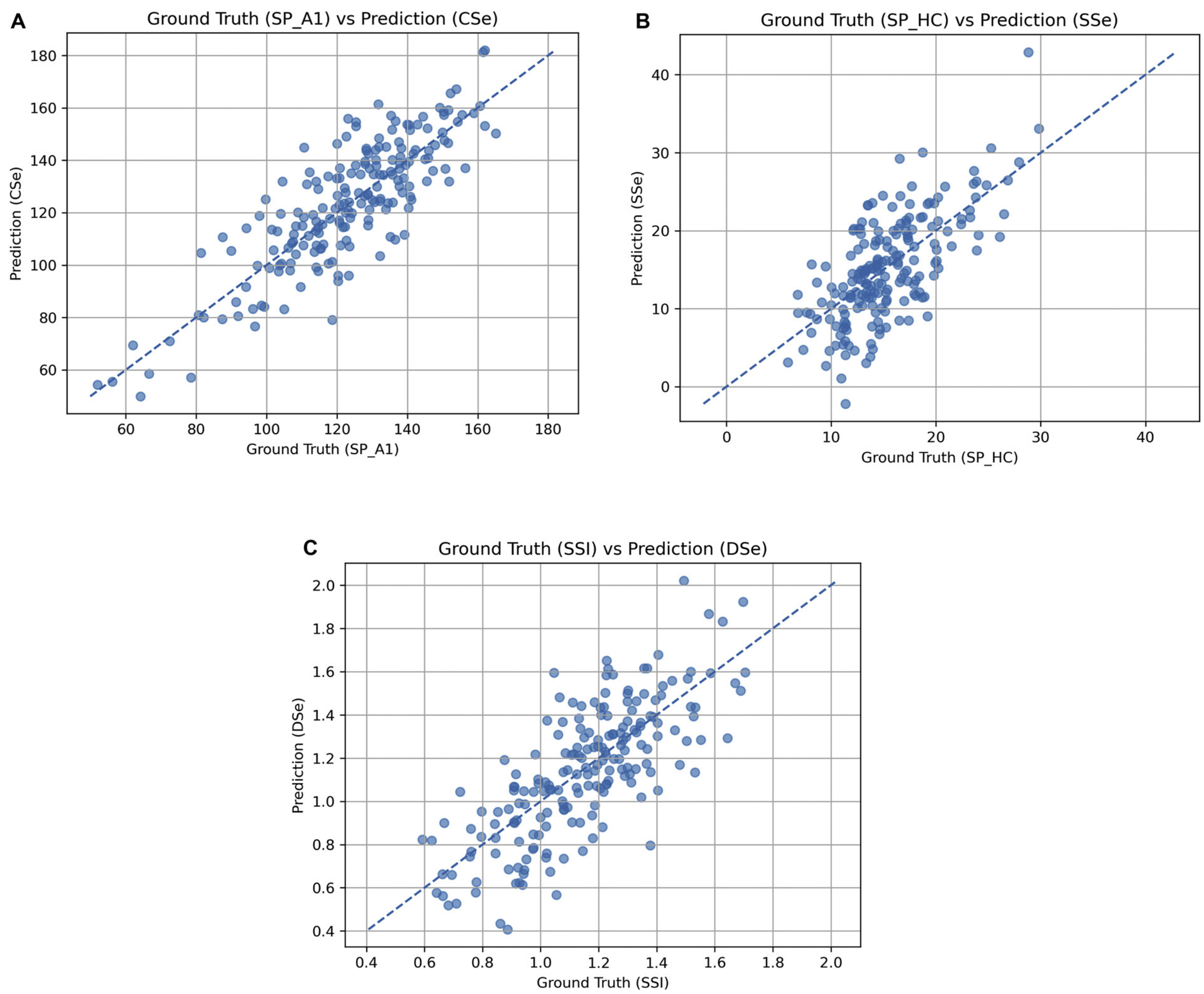

FIGURE 4. A. Predicted Ocular Response Analyzer (ORA) corneal stiffness estimate (CSe) versus stiffness parameter at first applanation (SP-A1) from Corvis ST of the development testing split, which is 20% of the development dataset. Dashed line is X = Y unity line. B. Predicted ORA scleral stiffness estimate (SSe) versus stiffness parameter at highest concavity (SP-HC) from Corvis ST of the development testing split which is 20% of the development dataset. Dashed line is X = Y. C. Predicted ORA deformation stiffness estimate (DSe) versus stress–strain index (SSI) from Corvis ST of the development testing split which is 20% of the development dataset. Dashed line is X = Y.

TABLE 5. Stiffness Prediction of the Testing Cohort Using AT-MoE.

	MSE (95% CI)	RMSE (95% CI)	MAE (95% CI)
Corneal stiffness estimate	0.0065 (0.0053-0.0078)	0.0806 (0.0729-0.0885)	0.0583 (0.0519-0.0650)
Scleral stiffness estimate	0.0109 (0.0089-0.0131)	0.1044 (0.0942-0.1144)	0.0864 (0.0773-0.0957)
Deformation stiffness estimate	0.0188 (0.0152-0.0225)	0.1371 (0.1232-0.1501)	0.0975 (0.0875-0.1087)

AT-MoE = Adaptive Temporal Mixture of Experts; CI = confidence interval; MAE = mean absolute error; MSE = mean squared error; RMSE = root mean squared error.

TABLE 6. Ocular Response Analyzer Stiffness Estimates in Independent Validation Dataset.

Validation Cohorts	Corneal Stiffness Estimate	Scleral Stiffness Estimate	Deformation Stiffness Estimate	KMI
Normal controls	126 ± 27	13.1 ± 3.4	1.3 ± 0.3	1.02 ± 0.31
Keratoconus	87 ± 29	9.0 ± 3.0	0.9 ± 0.2	0.38 ± 0.46

KMI = Keratoconus Match Index.

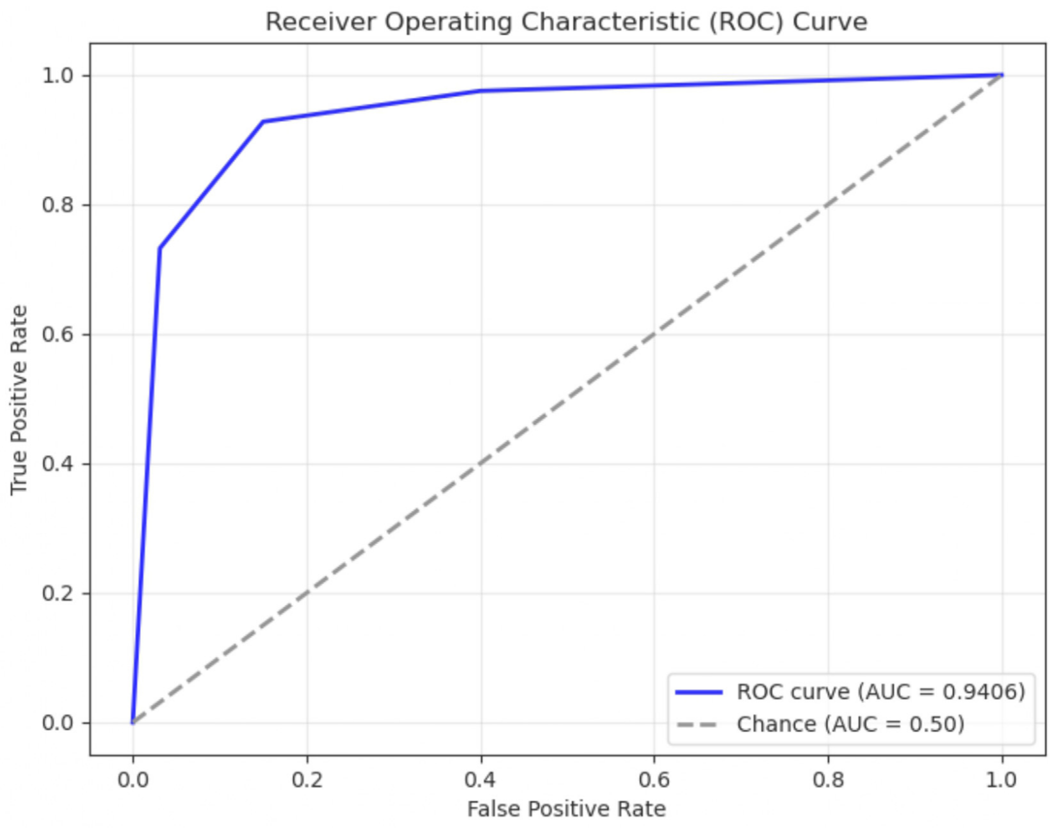

FIGURE 5. Receiver operating characteristic (ROC) curve for long short-term memory (LSTM) model prediction of keratoconus vs normal subjects in independent validation dataset. AUROC = 0.9406.

The predicted ORA Stiffness Estimates are given in Table 6 for the independent validation dataset. These estimates were used for the classification task of keratoconus detection, and also benefited from the AT-MoE architecture. As shown in Figures 5 and 6, the AUROC significantly improved from 0.9406 for the single LSTM model to 0.9512 for AT-MoE, with P = .0256 and CI = 0.0013, 0.0199 for △ AUROC = 0.0106. This indicates that AT-MoE enhances the model's discriminative power in distinguishing keratoconus from normal cases, reinforcing the benefit of modular expert design and dynamic routing in capturing clinically relevant temporal patterns. This can be compared to the performance of the Keratoconus Match Index (KMI), which uses only the waveform parameters to biomechanically discriminate between keratoconus and normal corneas. For the current dataset, the AUROC for KMI was 0.8771, which was significantly lower than LSTM with P = .008 and CI = 0.0166, 0.1105 for △ AUROC = 0.0636, and significantly lower than AT-MoE with P = .0021 and CI = 0.0269, 0.1215 for △ AUROC = 0.0742, highlighting the advantage of a machine learning approach that includes the pressure and applanation signals, along with the waveform parameters. The regression analysis of CSe versus Cspot of 77 eyes from 43 keratoconic subjects who had both ORA and topographic measurements, shows a statistically significant but weak, negative relationship (P = .0072; R^2 = 0.09) with greater curvature associated with lower corneal stiffness estimate, as expected. This is shown in Figure 7.

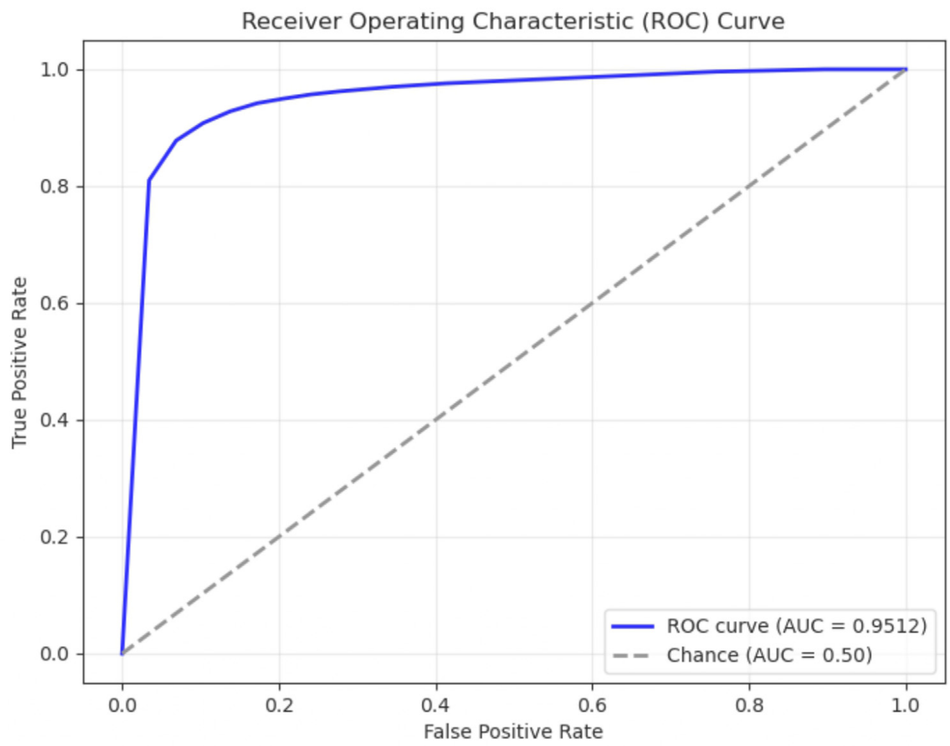

FIGURE 6. Receiver operating characteristic (ROC) curve for Adaptive Temporal Mixture of Experts (AT-MoE) model prediction of keratoconus vs normal subjects in independent validation dataset. AUROC = 0.9512.

DISCUSSION

The low-error estimation of elastic stiffness metrics from the ORA pressure and applanation waveforms using a sophisticated machine learning approach confirms the similarity of response of the viscoelastic cornea to air-puff deformation, despite differences in air-puff strategy and mechanism of quantifying response between the ORA and Corvis ST. However, these differences make viscoelastic parameters more easily determined by the ORA where the cornea never reaches its limit of deformation to the customized magnitude air-puff, so the first and second applanation pressures, corresponding to loading and unloading pressures, are almost purely corneal. Elastic stiffness parameters are more easily determined by the Corvis ST because, once the cornea reaches its limit of deformation, which is a clean endpoint for loading, the consistent air-puff magnitude continues to increase, and the excess energy of the air-puff generates backward whole-eye motion. Therefore, the second applanation pressure during unloading in the Corvis ST is distributed to both cornea and whole-eye motion, requiring a complicated mathematical approach to isolate the corneal component.[32]

From a clinical perspective, this study represents a major advancement in the performance of the ORA. Currently, the only biomechanical response metrics available for analysis to clinicians in the United States are viscoelastic in nature, including CH. Although CH is a useful indicator for the conversion of ocular hypertension to open-angle glaucoma[33] and for the development of progressive structural and functional loss in patients with preexisting open-angle glaucoma,[34] its utility for the detection of keratoconus is limited, as discussed previously. This limitation was originally addressed through the development of the Keratoconus Match Index (KMI) for the ORA. The KMI is a set of 7 waveform parameters, which were derived from an analysis of keratoconus and control subjects from three continents: France, the United States, and Brazil. Waveform parameters were compared in a proprietary manner to determine the subset of waveform parameters that best differentiated the two groups. Even though Labris et al. used the KMI to differentiate patients with keratoconus,[35,36] even subclinical disease, from healthy controls with accuracy greater than CH alone, the KMI metric never gained traction among clinicians and was not included in the newest generation of the device, generation 3 (G3), because of changes to the air-puff delivery system, working distance, and applanation detection. A recent reassessment of the diagnostic accuracy of the KMI for keratoconus yielded only moderate accuracy,[31] possibly due to high collinearity between ORA waveform parameters. In comparing KMI to the ORA stiffness estimates in the current study, the KMI showed an AUROC of only

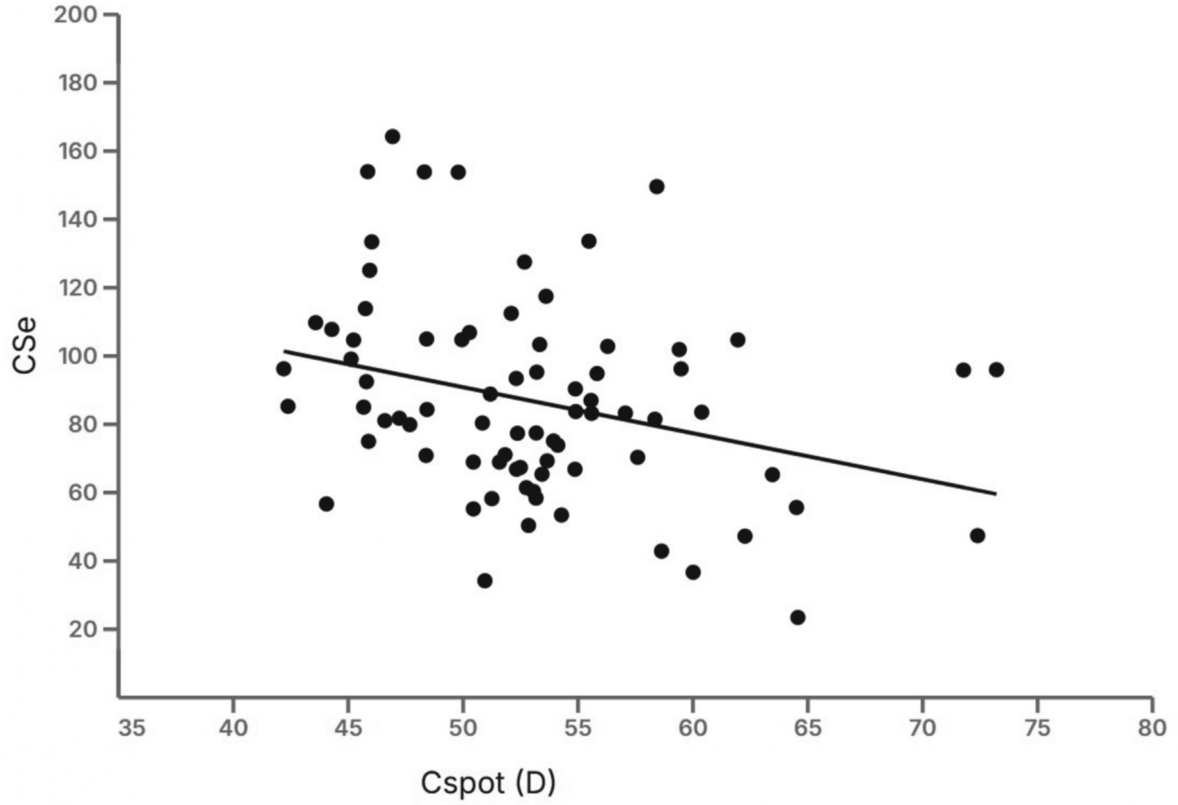

FIGURE 7. Linear regression analysis of predicted Ocular Response Analyzer (ORA) corneal stiffness estimate (CSe) versus maximum curvature in 2-mm diameter circle on anterior tangential topography (Cspot in diopters [D]), showing significant negative relationship ($P = 0.0072$; $R^2 = 0.09$) indicating that greater maximum curvature (increasing severity of keratoconus) is associated with lower estimated corneal stiffness.

0.8771, whereas the current machine modeling approaches increased the AUROC to 0.9512 using the same dataset. This highlights the importance of information in the pressure and applanation signals, beyond the discrete wavform parameters.

The novel AT-MoE model expands the utility by predicting new metrics for stiffness, where the stiffness parameters of the Corvis ST serve as Ground Truth, and a subset of the waveform parameters along with the raw pressure-applanation waveforms themselves are used to parse eyes with keratoconus from healthy eyes. A detailed comparison of AT-MoE and LSTM is provided in the Supplemental Material. In addition to aiding in the detection of keratoconus, this machine learning model has the potential to give clinicians and researchers in the United States access to ocular stiffness metrics that were previously inaccessible, opening new avenues for research and discovery in keratoconus, glaucoma, and other ocular diseases. The next steps in this line of research are to build the validation database of this model by expanding the sample of healthy and eyes with keratoconus, especially eyes with subclinical disease. Detection of keratoconus in its earliest stages will allow for timely interventions to slow or to arrest progression in order to preserve vision. Additionally, the integration of corneal tomography and/or topography images into the model will likely improve its ability to detect keratoconus beyond its current capabilities by including an assessment of asymmetry, which is a hallmark of keratoconus. Previous diagnostic tools that use ocular biomechanics and corneal imaging techniques have shown very high accuracy at detecting keratoconus.[6,37] The problem is that these tools are dependent on elastic stiffness parameters from the Corvis ST, which has limited their application in eye clinics in the United States. The advancements made here have the potential to solve this issue by allowing for the generation of elastic stiffness metrics from the ORA, which is more ubiquitous in the United States than the Corvis ST.

There are several limitations to this study. The independent validation dataset is ORA only, without Corvis ST parameters to provide Ground Truth comparison of specific parameter estimates. However, the overall goal is to use the ORA in a biomechanical assessment for the detection of keratoconus. Our model separated the keratoconus subjects from the controls with an AUROC of over 0.95, which is a performance close to published algorithms using tomography. In addition, CCT was not acquired in the existing validation dataset to allow staging of keratoconus beyond evaluation of maximum curvature. However, CCT was inherent to the deformation response of the cornea, so it contributed to the stiffness estimates by the model. In addition, the individual maximum curvature values are shown in Figure 7, where only three values are above 70 D. All other values are below 65 D, with most below 55D. The Cspot distribution is skewed toward lower values of maximum curvature, suggesting less advanced disease, contributing to the generalizability of the model.

In conclusion, this work presents the Adaptive Temporal Mixture of Experts (AT-MoE) model as a powerful framework for modeling complex sequential data in optometry and ophthalmology. By combining the temporal modeling strengths of LSTMs with dynamic gating and expert modularization, AT-MoE effectively captures both short- and long-term biomechanical trends in ocular waveform data. Experimental results on multi-cohort clinical datasets show that AT-MoE significantly improves prediction accuracy for corneal, scleral, and deformation stiffness compared to conventional LSTM models. Furthermore, the model enhances keratoconus classification performance, not only over the LSTM model but also over the original KMI, highlighting its clinical relevance. With its ability to adaptively route inputs to specialized experts, the AT-MoE framework provides a scalable and interpretable solution for future applications in ocular disease detection and personalized diagnostics.

Generation of elastic stiffness metrics from the ORA is novel and has the potential to enhance its clinical utility by giving clinicians in the United States new access to biomechanical metrics beyond CH. Moreover, deployment of this model in primary care eye clinics may foster earlier detection of keratoconus using a simple air-puff test than what is currently available. This new technology may be further enhanced by the addition of corneal and anterior segment imaging modalities already in place.

CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

Yue Zhang: Writing – original draft, Software, Methodology, Conceptualization. **Swati Padhee:** Software, Writing – review & editing. **Phillip T. Yuhas:** Writing – review & editing, Supervision, Resources, Data curation. **Cynthia J. Roberts:** Writing – review & editing, Supervision, Resources, Funding acquisition, Data curation, Conceptualization. **Srinivasan Parthasarathy:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Funding/Support: Funding was provided by NIH R01EY027399 Corneal Biomechanics in Ocular Disease; NIH P30EY032857 Vision Sciences Research Core Program; Ohio Lions Eye Research Foundation W. R. Bryan Diabetic Eye Disease Fund Research to Prevent Blindness (RPB) New Chair Challenge Grant.
Financial Disclosures: P.Y. received travel support from Reichart in the past (no influence on the methodology or results reported in the current manuscript). C.J.R. is a consultant to Ziemer Ophthalmic Systems AG and OCULUS Optikgeräte GmbH. All other authors report no financial disclosures or conflicts of interest. All authors attest that they meet the current ICMJE criteria for authorship.
Acknowledgments: The authors appreciate the careful data acquisition of Clinical Research Coordinator Hayley Griggs, and the Department of Ophthalmology & Visual Sciences Clinical Research Team.

REFERENCES

1. Santodomingo-Rubido J, Carracedo G, Suzaki A, Villa-Collar C, Vincent SJ, Wolffsohn JS. Keratoconus: an updated review. *Cont Lens Anterior Eye*. 2022;45:101559.
2. Weinberger NM, Diamond DM. Physiological plasticity in auditory cortex: rapid induction by learning. *Prog Neurobiol*. 1987;29:1–55.
3. Ferdi AC, Kandel H, Nguyen V, et al. Five-year corneal cross-linking outcomes: a Save Sight Keratoconus Registry Study. *Clin Exp Ophthalmol*. 2023;51:9–18.
4. Roberts CJ, Dupps Jr WJ. Biomechanics of corneal ectasia and biomechanical treatments. *Cataract Refract Surg*. 2014;40:991–998.
5. Luz A, Lopes B, Hallahan KM, et al. Enhanced combined tomography and biomechanics data for distinguishing forme fruste keratoconus. *J Refract Surg*. 2016;32:479–494.
6. Ambrósio Jr R, Machado AP, Leão E, et al. Optimized artificial intelligence for enhanced ectasia detection using Scheimpflug-based corneal tomography and biomechanical data. *Am J Ophthalmol*. 2023;251:126–142.
7. Vinciguerra R, Ambrósio Jr R, Elsheikh A, et al. Detection of keratoconus with a new biomechanical index. *J Refract Surg*. 2016;32:803–810.
8. Luce DA. Determining in vivo biomechanical properties of the cornea with an ocular response analyzer. *J Cataract Refract Surg*. 2005;31:156–162.
9. Fontes BM, Ambrósio Jr R, Velarde GC, Nosé W. Ocular response analyzer measurements in keratoconus with normal central corneal thickness compared with matched normal control eyes. *J Refract Surg*. 2011;27:209–215.
10. Qin X, Yu M, Zhang H, Chen X, Li L. The mechanical interpretation of ocular response analyzer parameters. *BioMed Res Int*. 2019;2019:5701236.
11. Aoki S, Murata H, Matsuura M, et al. The relationship between the waveform parameters from the ocular response analyzer and the progression of glaucoma. *Ophthalmol Glaucoma*. 2018;1:123–131.
12. Yuhas PT, Fortman MM, Mahmoud AM, Roberts CJ. Keratoconus cone location influences ocular biomechanical parameters measured by the Ocular Response Analyzer. *Eye Vis*. 2024;11:2.

13. Graves A. Chapter 3: Long short-term memory. *Supervised sequence labelling with recurrent neural networks*. Berlin: Springer; 2012:37–45.
14. Zhou M, Zhang Y, Karimi Monsefi A, et al. Reducing manual labeling requirements and improved retinal ganglion cell identification in 3D AO-OCT volumes using semi-supervised learning. *Biomed Opt Express*. 2024;15:4540–4556.
15. Zhou M, Zhang Y, Kirkendall E, et al. ISOSNet: a unified framework for cone photoreceptor detection and inner segment and outer segment length measurement from AO-OCT B-Scans. *Biomed Opt Express*. 2025;16:3237–3254.
16. Salarian S, Zhang Y, Padhee S, Parthasarathy S. MedEqualizer: a framework investigating bias in synthetic medical data and mitigation via augmentation [arXiv preprint]. arXiv:2511.01054. 2025.
17. Zhang Y, Roberts CJ, Padhee S, Doble N, Parthasarathy S, Yuhas PT. Machine learning detection of keratoconus through ocular response analyzer waveform parameters. *Invest Ophthalmol Vis Sci*. 2025;66 3378-3378.
18. Marsolo K, Parthasarathy S, Twa M, Bullimore MA. A model-based approach to visualizing classification decisions for patient diagnosis. *Conference on Artificial Intelligence in Europe* Springer; 2005:473–483.
19. Marsolo K, Parthasarathy S, Twa M, Bullimore MA. Classification of biomedical data through model-based spatial averaging. *Fifth IEEE Symposium on Bioinformatics and Bioengineering* IEEE; 2005:49–56.
20. Marsolo K, Twa M, Bullimore MA, Parthasarathy S. Spatial modeling and classification of corneal shape. *IEEE Transact Inform Technol Biomed*. 2007;11:203–212.
21. Twa MD, Parthasarathy S, Raasch TW, Bullimore M. Decision tree classification of spatial data patterns from videokeratography using Zernike polynomials. *Proceedings of the 2003 SIAM International Conference on Data Mining* SIAM; 2003:3–12.
22. Moshirfar M, Edmonds JN, Behunin NL, Christiansen SM. Corneal biomechanics in iatrogenic ectasia and keratoconus: a review of the literature. *Oman J Ophthalmol*. 2013;6:12–17.
23. Kondapalli S, Roberts C, Mahmoud A, Weber P, Peterson J. The effect of diabetes on biomechanical properties of the cornea. *Invest Ophthalmol Vis Sci*. 2010;51:4631.
24. Wang X, Xu G, Wang W, et al. Changes in corneal biomechanics in patients with diabetes mellitus: systematic review and meta-analysis. *Acta Diabetol*. 2020;57:973–981.
25. McMonnies CW. The interaction between intracranial pressure, intraocular pressure and lamina cribosal compression in glaucoma. *Clin Exp Optom*. 2016;99(3):219–226.
26. Scott JA, Roberts CJ, Mahmoud AM, Jain SG. Evaluating the relationship of intraocular pressure and anterior chamber volume with use of prostaglandin analogues. *J Glaucoma*. 2021;30:421–427.
27. Nguyen BA, Reilly MA, Roberts CJ. Biomechanical contribution of the sclera to dynamic corneal response in air-puff induced deformation in human donor eyes. *Exp Eye Res*. 2020;191:107904.
28. Eliasy A, Chen KJ, Vinciguerra R, et al. Determination of corneal biomechanical behavior in-vivo for healthy eyes using CorVis ST tonometry: stress-strain index. *Front Bioeng Biotechnol*. 2019;7:105.
29. Roberts CJ, Mahmoud AM, Bons JP, et al. Introduction of two novel stiffness parameters and interpretation of puff-induced biomechanical deformation parameters with a dynamic Scheimpflug analyzer. *J Refract Surg*. 2017; 33:266–273.
30. Yuhas PT, Fortman MM, Nye M, Mahmoud AM, Roberts CJ. Waveform score influences the outcome metrics of the ocular response analyzer in patients with keratoconus and in healthy controls. *Curr Eye Res*. 2025;50(7):700–709.
31. Yuhas PT, Fortman MM, Nye M, Mahmoud AM, Roberts CJ. Intraocular pressure and ocular biomechanical parameters vary between generations of the ocular response analyzer in healthy and ectatic eyes. *Front Med*. 2025;12:1605641.
32. Roberts CJ. Concepts and misconceptions in corneal biomechanics. *J Cataract Refract Surg*. 2014;40:862–869.
33. Pensyl D, Sullivan-Mee M, Torres-Monte M, Halverson K, Qualls C. Combining corneal hysteresis with central corneal thickness and intraocular pressure for glaucoma risk assessment. *Eye*. 2012;26:1349–1356.
34. Chung E, Demetriades AM, Christos PJ, Radcliffe NM. Structural glaucomatous progression before and after occurrence of an optic disk haemorrhage. *Br J Ophthalmol*. 2015;99:21–25.
35. Labiris G, Gatzioufas Z, Sideroudi H, Giarmoukakis A, Kozobolis V, Seitz B. Biomechanical diagnosis of keratoconus: evaluation of the keratoconus match index and the keratoconus match probability. *Acta Ophthalmol*. 2013;91:e258–e262.
36. Labiris G, Giarmoukakis A, Gatzioufas Z, Sideroudi H, Kozobolis V, Seitz B. Diagnostic capacity of the keratoconus match index and keratoconus match probability in subclinical keratoconus. *J Cataract Refract Surg*. 2014;40:999–1005.
37. Xian Y, Zhao Y, Sun L, et al. Comparison of bilateral differential characteristics of corneal biomechanics between keratoconus and normal eyes. *Front Bioeng Biotechnol*. 2023;11:1163223.